

From Residual Krylov Geometry to Frozen Production Bounds

A Ten-Layer Audit, Certificate-Stack Closure, and the Two-Gate Stage A1 Frontier

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July 2026

Abstract

RH-62–RH-70 pursued the directional Hardy gate in the post-RH-50 roadmap. The tools were residual Krylov identities, weighted metrics, block cross-column fusion, covariance-adapted envelopes, phase-coherence lower bounds, adaptive certificate selection, and finally outward-rounded production execution. This paper is the tenth synthesis layer.

We formulate the accumulated mechanisms as a certificate stack. Let $K = (K_m)_{m \geq 0}$ be the true transfer sequence and $\hat{K} = (\hat{K}_m)_{m \geq 0}$ a frozen finite-matrix sequence in the Hilbert space $\ell^2(\mathbb{N}_0; \mathbb{C}^{t \times s})$. If terminal candidate c proves $\|\hat{K}\|_{\mathcal{H}} \leq U_c$ and an upstream bridge proves $\|K - \hat{K}\|_{\mathcal{H}} \leq \Delta$, then

$$\|K\|_{\mathcal{H}} \leq \min_c U_c + \Delta.$$

A candidate-independent bridge preserves the terminal Pareto frontier. If a certified finite-prefix lower is F^- and a frozen full upper is U^+ , then

$$\Delta \leq (1 + \tau)F^- - U^+$$

is sufficient for a total relative target τ . Polylogarithmic terminal and bridge bounds compose into a polylogarithmic true-family bound.

A 256-bit Arb audit applies this slack formula to the ten RH-70 left/right balls. Every row leaves positive bridge headroom inside a one-percent total target. The minimum certified absolute slack is $8.41145551828 \times 10^{-4}$, and the minimum relative slack is $6.30626677513 \times 10^{-4}$, or 0.0630627% of the finite prefix. This one-percent target is an engineering sharpness ledger, not a Stage A1 necessity.

The dependency audit makes a crucial distinction. The first-open frontier for a finite-scale end-to-end Hardy certificate is the single upstream interval enclosure of the folded-Gaussian, rank-two deflation, source, and observation triple. Full Stage A1 has two open gates: that upstream inclusion and a uniform small-noise/dyadic family bound. Thus the terminal arithmetic wall is removed, but asymptotic uniformity is not.

The route remains viable, while three shortcuts are now closed in their stated forms: plain norm propagation, unlocalized full-space Lyapunov weighting, and universal fixed Krylov depth. Stages A5–D—the pole-renormalized determinant, canonical scattering completion, self-adjoint generator, $T \log T$ law, prime-power trace formula, completed zeta identity, and Riemann Hypothesis implication—remain unreached.

Keywords: certificate stack; Krylov residual; Hardy energy; interval arithmetic; dependency frontier; small-noise spectral limit.

MSC 2020: 47A10; 47B65; 65G20; 65F35; 90C29; 93B28.

1 Introduction

The post-RH-50 roadmap placed the program inside Stage A: construct a canonical small-noise intrinsic determinant before discussing scattering, self-adjointness, counting, or arithmetic trace identities [12]. The immediate obstruction was a pair of directional Hardy energies. Subsequent

work closed the sufficient peripheral-factor transfer, the analytic adaptive cutoff transfer, and a conditional intrinsic-identification composition theorem [6, 8, 13]. The uniform Hardy input itself remained open.

RH-62–RH-70 then followed the tail mechanism through nine local layers. The exploration was not monotone: some layers strengthened an upper, some identified a necessary cost, and some proved exact no-go results. RH-70 finally executed a complete outward-rounded terminal audit on the frozen production matrices [7].

This review has two purposes. First, it extracts a theorem-level composition from the local mechanisms. Second, it prevents the statement that RH-70 is green from being overextended. A frozen terminal certificate and an end-to-end physical-family theorem are different statements.

Main conclusions

1. Terminal candidate selection and an upstream transfer-difference certificate form a simple, rigorous Hilbert-space stack.
2. The RH-70 balls leave certified finite-scale room for such a bridge, although the one-percent room is narrow at two middle scales.
3. The finite-scale open frontier has one gate; Stage A1 has two.
4. Additional terminal-tail variants are lower priority than upstream inclusion and uniform family scaling.
5. The Hilbert–Polya-oriented Stages B–D remain logically untouched.

2 Inherited stage architecture

The relevant portion of the roadmap is

A1: directional Hardy family bound $\longrightarrow$ A4: intrinsic Riesz identification
 $\longrightarrow$ A5: pole-renormalized determinant limit.

The numbering retains the intervening factor and cutoff gates. RH-52 closed a sufficient range-restricted peripheral transfer rather than the stronger factor norm originally requested. RH-55 closed the analytic strong–weak adaptive cutoff transfer. RH-54 proved the conditional exponent ledger

$$\|\mathcal{I}_{n,\sigma}\|_{S_2} = O\left(n^{-2}\sigma^{-13/4-\alpha_B-\alpha_C}\right),$$

with every strict $n\sigma^2 \rightarrow \infty$ schedule preserved when $\alpha_B + \alpha_C \leq 1/4$.

Accordingly, the active mathematical input is still a controlled left/right Hardy family. A finite list of bounded energies suggests such control but does not prove it.

3 Certificate-stack closure

Let

$$\mathcal{H} = \ell^2(\mathbb{N}_0; \mathbb{C}^{t \times s}), \quad \|K\|_{\mathcal{H}}^2 = \sum_{m \geq 0} \|K_m\|_{\mathbb{F}}^2.$$

For a physical or analytic transfer triple (A, X, Y) , the coefficients are $K_m = Y A^m X$. Let $\hat{K}$ denote coefficients from one frozen finite-matrix triple.

At a fixed scale, terminal candidate c carries a valid upper U_c for $\|\hat{K}\|_{\mathcal{H}}$ and a nonnegative cost vector b_c . RH-69 formalized Pareto pruning of these candidates [11].

Theorem 3.1 (Certificate-stack closure). *Suppose*

$$\left\| \hat{K} \right\|_{\mathcal{H}} \leq U_c \quad (c \in \mathcal{C}), \quad \left\| K - \hat{K} \right\|_{\mathcal{H}} \leq \Delta.$$

Then:

(i)

$$\|K\|_{\mathcal{H}} \leq \min_{c \in \mathcal{C}} U_c + \Delta.$$

(ii) *If Δ is candidate independent, replacing each U_c by $U_c + \Delta$ leaves every upper-and-cost Pareto dominance relation, and hence the Pareto frontier, unchanged.*

(iii) *Let F be the frozen finite-prefix norm. If interval data give $F^- \leq F$ and $\min_c U_c \leq U^+$, then*

$$\Delta \leq (1 + \tau)F^- - U^+ \tag{1}$$

is sufficient for $\|K\|_{\mathcal{H}} \leq (1 + \tau)F$ whenever the right side is nonnegative.

(iv) *If $\min_c U_c = O(\log(1/\sigma)^a)$ and $\Delta = O(\log(1/\sigma)^b)$, then*

$$\|K\|_{\mathcal{H}} = O\left(\log(1/\sigma)^{\max(a,b)}\right).$$

Proof. The triangle inequality in $\mathcal{H}$ gives

$$\|K\|_{\mathcal{H}} \leq \left\| \hat{K} \right\|_{\mathcal{H}} + \left\| K - \hat{K} \right\|_{\mathcal{H}} \leq U_c + \Delta$$

for every c ; minimizing proves (i). Adding the same scalar to every upper does not change any pairwise upper inequality, so all dominance relations are preserved, proving (ii). Under (1),

$$\|K\|_{\mathcal{H}} \leq U^+ + \Delta \leq (1 + \tau)F^- \leq (1 + \tau)F,$$

which is (iii). The sum of two polylogarithmic quantities has the larger of the two exponents, proving (iv). $\square$

Remark 3.2 (Candidate-dependent bridges). *If the bridge changes with the terminal representation, candidate c must carry total upper $U_c + \Delta_c$ and any new bridge cost. Pareto pruning is then rerun on the augmented candidate; the common-bridge invariance does not apply.*

Remark 3.3 (The role of the one-percent target). *Equation (1) is useful for designing a sharp finite-scale validation. Stage A1 itself asks for an asymptotic upper compatible with the RH-54 exponent budget. It does not require one-percent agreement with a frozen diagnostic.*

4 Audit of the nine input layers

The nine archived summaries contain forty boolean theorem gates. Every gate is true in its stated scope. These flags are an archive-consistency check, not a count of independent theorems.

Table 1: RH-62–RH-70 route audit. RH-71 is the tenth synthesis layer.

RH	Durable result	Route effect	Boundary exposed
62	Arnoldi residual identity and directional power upper	advance	residual propagation remained nondirectional
63	coherent nested closure and breakdown exactness	advance	uniform recursion depth remained open
64	Lyapunov-weighted terminal contraction	advance	metric conditioning became the cost
65	sharp Jordan coupling/gap threshold	no-go boundary	global weighting cannot replace localization
66	block Galerkin and cross-column Gram certificates	advance	uniform PSD majorization can lose physical sharpness
67	covariance-optimal allocation and sharpness/size duality	advance/correct	focusing may force global PSD growth
68	Fourier-ring fixed-depth obstruction	exact no-go	phase compression or growing depth is necessary
69	Pareto portfolio and three-way triage	synthesis	production interval execution remained absent
70	exact-dyadic production block-Hardy audit	validated advance	upstream inclusion and uniformity remain open

4.1 Positive chain

RH-62 and RH-63 replaced a worst-direction norm envelope by exact residual identities and coherent recursive closure [9, 10]. RH-64 moved the terminal remainder into a positive Lyapunov metric [14]. RH-66 then preserved cancellation across source columns with a block Galerkin identity and coefficient-aware Grams [2]. RH-67 optimized positive residual allocations against a physical covariance [4]. RH-69 assembled the incomparable mechanisms without turning failed uppers into no-go claims. RH-70 supplied the first production-scale outward-rounded terminal execution in this chain.

4.2 No-go results and correction

RH-65 proves that canonical full-space Lyapunov weighting can become polynomially or super-polynomially ill-conditioned when nilpotent coupling exceeds the peripheral gap scale [3]. This closes a global shortcut, not localized residual weighting.

RH-68 proves that perfect stability and metric conditioning do not imply a universal fixed block depth: normalized Krylov vectors can be a Fourier basis [5]. This closes fixed depth from stability data alone, not physical phase compression or a controlled growing depth.

RH-67 also corrected the RH-66 cancellation diagnostic. Forming a large Gram in the original coefficient coordinates produced the archived binary64 gain 410.16; factor-first formation and exact scalar reduction give the mathematical isotropic gain 2.92454. The RH-66 identities and upper theorems were unchanged. The correction is part of the durable record, not an inconvenient outlier.

5 Certified bridge-slack ledger

RH-70 archives Arb balls for the frozen finite-prefix norm F and frozen full upper U . We reparsed those balls at 256-bit precision [1] and evaluated

$$S_\tau = (1 + \tau)F - U$$

with outward rounding. Positive lower endpoint certifies that an upstream bridge of that size can be inserted while retaining the total target.

Table 2: Certified one-percent bridge slack as a percentage of the frozen finite prefix.

σ	horizon	left slack (%)	right slack (%)
0.16	4	0.643334	0.678671
0.08	9	0.600404	0.689896
0.04	16	0.063063	0.088419
0.02	25	0.076311	0.219805
0.01	32	0.143406	0.205715

All ten lower endpoints are positive. The minimum occurs at $\sigma = 0.04$ on the left:

$$S_{0.01} \geq 0.0008411455518283064, \quad \frac{S_{0.01}}{F} \geq 0.0006306266775125892.$$

For a two-percent total target, the minimum relative bridge budget becomes 1.0630627%; for a five-percent target it becomes 4.0630627%.

The calculation proves headroom, not the bridge. It also shows why target selection matters. A one-percent end-to-end recreation would require a quite sharp upstream enclosure at the middle scales, whereas the asymptotic Stage A1 theorem may tolerate a larger controlled term.

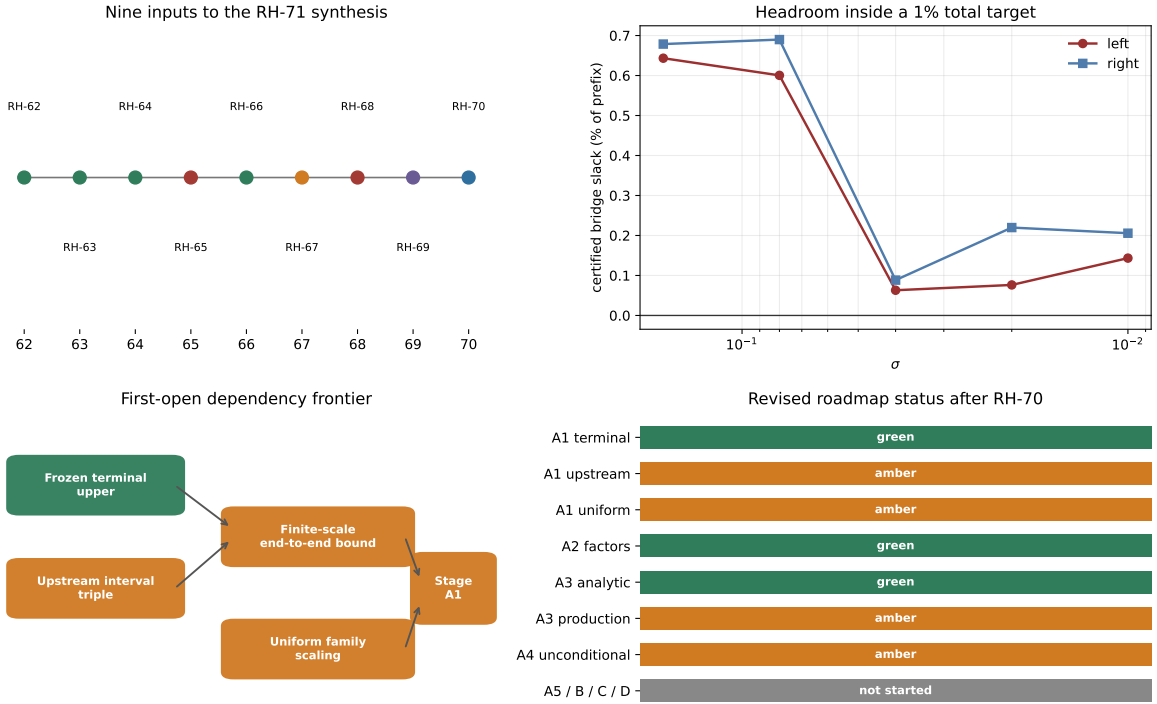


Figure 1: The nine input layers, certified bridge headroom, first-open dependency frontier, and revised roadmap status. Red timeline points are route-closing no-go results, not failures of all directional methods.

6 First-open dependency frontiers

To avoid conflating a finite audit with a family theorem, we encode the remaining claims as a finite acyclic prerequisite graph.

Definition 6.1 (First-open frontier). Let $\mathcal{G}$ be a finite acyclic graph whose nodes close monotonically once all named prerequisites close. Given a closed set $\mathcal{C}$ and a target v , recursively descend through every nonclosed prerequisite. If a nonclosed node has no nonclosed prerequisite, retain it. The resulting antichain is the first-open frontier $\mathfrak{F}(v; \mathcal{C})$.

Proposition 6.2 (Frontier decomposition). *Relative to a fixed prerequisite graph:*

- (i) *every unresolved derivation branch from v reaches a node of $\mathfrak{F}(v; \mathcal{C})$;*
- (ii) *closing all nodes of the frontier permits closure to propagate upward until either v closes or a new first-open frontier is exposed;*
- (iii) *no proof following only the named dependency edges can close v while leaving a current frontier node open.*

Proof. Topologically order the finite graph. Recursive descent must terminate at a closed node or a first nonclosed node with no nonclosed prerequisite, proving (i). Upward induction in the reverse topological order proves (ii). Statement (iii) is the contrapositive of the monotone prerequisite rule. $\square$

This proposition is bookkeeping relative to the declared graph; it does not prove that no future theorem can bypass a named node. Its value is claim hygiene.

The current graph contains the closed terminal frozen upper and the closed augmented difference identity from RH-70. The difference *instance* still requires an interval-enclosed upstream triple. A finite-scale end-to-end bound requires the terminal upper and that bridge. Stage A1 then additionally requires a uniform family-scaling theorem.

Corollary 6.3 (Two distinct frontiers). *For the archived dependency graph,*

$$\mathfrak{F}(\text{finite-scale Hardy}) = \{\text{upstream interval triple}\},$$

whereas

$$\mathfrak{F}(\text{Stage A1}) = \{\text{upstream interval triple}, \text{uniform family scaling}\}.$$

The same two nodes are inherited by unconditional Stage A4 because the peripheral-factor transfer, adaptive cutoff transfer, and conditional identification composition theorems are already green in their stated scopes. The renormalized determinant target has one additional future node: the pole-renormalized limit theorem itself.

7 Revised roadmap after RH-70

Table 3: Stage status after the ten-layer block.

Gate	Status	Meaning
A1 terminal finite-matrix mechanism	green	exact identities and frozen interval execution
A1 upstream production inclusion	amber	matrix construction, deflation, and transfer not enclosed
A1 uniform small-noise family bound	amber	no analytic dyadic/polylog theorem
A2 sufficient peripheral transfer	green	range-restricted gate closed by RH-52
A3 analytic adaptive cutoff transfer	green	strong–weak transfer closed by RH-55
A3 production interval composition	amber	shares the upstream inclusion gate
A4 conditional composition	green	RH-54 exponent ledger
A4 unconditional identification	amber	inherits the two A1 gates
A5 pole-renormalized determinant	not started	waits for unconditional A4
Stages B–D	not started	no canonical scattering, self-adjoint, counting, or arithmetic object

7.1 Immediate priority

The next productive target is an upstream interval triple, not another terminal-tail formula:

1. enclose folded-Gaussian row construction and normalization;
2. exploit the exact row-stochastic Perron right vector;
3. validate the stationary left vector and parity Riesz pair by residual, primal–dual, or contour methods;
4. enclose the rank-two bulk subtraction and coarse/detail source and observation blocks;
5. execute the RH-70 augmented difference certificate.

A direct interval eigendecomposition is not compulsory. The contour, residual, and low-rank certification technology developed earlier in the series can validate only the two peripheral factors needed by the bulk subtraction.

7.2 Continuation, fallback, and stopping signals

Continue if the upstream bridge is finite with cost compatible with the RH-54 quarter-power ledger and if one Hardy radius survives on the intended dyadic schedules. The next analytic theorem must control horizon, phase support, or effective rank; RH-68 forbids assuming fixed depth from stability alone.

Fallbacks remain legitimate:

- relax the one-percent engineering target;
- use strong–weak analytic perturbation rather than a complete interval eigensolver;
- permit a proved growing block depth;
- use localized weighted residuals after deflation rather than a global metric.

A decisive stopping signal for the present Stage-A route would be a validated proof of no common Hardy radius, an unavoidable combined energy exponent greater than $1/4$ under every admissible mesh regime, or super-polynomial directional/upstream growth with no canonical renormalization. Missing the one-percent finite-scale target alone is not such a signal.

8 Position on the Hilbert–Polya-oriented route

The current result is still inside the small-noise transfer-determinant stage. It neither constructs nor suggests declaring the noisy Markov operator to be self-adjoint. Stage A5 is the pole-renormalized determinant limit; only after it closes do Stages B–D retain their original order:

1. close the pole-renormalized small-noise determinant;
2. construct a canonical order-sensitive scattering or inner object;
3. derive a self-adjoint realization and its intrinsic $T \log T$ counting law;
4. derive a target-independent prime-power trace formula;
5. prove a completed-zeta identity with multiplicities and no spurious or missing levels;
6. only then ask for a Riemann Hypothesis implication.

The independent TPC branch may contribute arithmetic techniques, but there is still no interface theorem preserving test functions, weights, multiplicities, normalization, and limiting errors. Shared terminology does not create such an interface.

9 Conclusion

The RH-62–RH-71 block has a coherent outcome. Residual identities, nested closure, localized weighting, block fusion, covariance adaptation, lower gates, portfolio selection, and exact-dyadic execution now form a composable certificate stack. The terminal frozen calculation survives outward rounding and leaves positive room for an upstream bridge on every archived row.

Two broad shortcuts are definitively too strong: global weighting and universal fixed depth. Their failure narrows rather than ends the route. The next finite-scale wall is upstream inclusion; the next asymptotic wall is uniform family scaling. Keeping those two gates separate is the main route correction of this review.

References

- [1] Fredrik Johansson. Arb: Efficient arbitrary-precision midpoint-radius interval arithmetic. *IEEE Transactions on Computers*, 66(8):1281–1292, 2017. doi: 10.1109/TC.2017.2690633.
- [2] Liang Wang. Block cross-column krylov gram certificates for phase-fused stein tails. Preprint and software repository, RH-66, 2026.
- [3] Liang Wang. Conditioning thresholds for lyapunov metrics on near-peripheral jordan families. Preprint and software repository, RH-65, 2026.
- [4] Liang Wang. Physical-covariance block residual envelopes. Preprint and software repository, RH-67, 2026.
- [5] Liang Wang. Phase-coherence barriers to uniform block krylov depth. Preprint and software repository, RH-68, 2026.
- [6] Liang Wang. Intrinsic peripheral residue transfer from weak finite factors. Preprint and software repository, RH-52, 2026.
- [7] Liang Wang. Outward-rounded block hardy certificates for frozen production matrices. Preprint and software repository, RH-70, 2026.
- [8] Liang Wang. Factor-aware sparse-to-full transfer for intrinsic riesz identification. Preprint and software repository, RH-54, 2026.
- [9] Liang Wang. Krylov residual certificates for directional stein tails. Preprint and software repository, RH-62, 2026.
- [10] Liang Wang. Nested arnoldi residual closure for directional stein tails. Preprint and software repository, RH-63, 2026.
- [11] Liang Wang. An adaptive upper–lower certificate portfolio for directional stein tails. Preprint and software repository, RH-69, 2026.
- [12] Liang Wang. Quadratic prime-dynamics roadmap after RH-50. Program audit and staged roadmap, 2026. URL https://github.com/maris205/prime_dynamics_theory/blob/main/papers/RH-ROADMAP-after-RH50.md.
- [13] Liang Wang. Strong–weak adaptive cutoff transfer for intrinsic riesz factors. Preprint and software repository, RH-55, 2026.
- [14] Liang Wang. Observability-weighted terminal residuals for nested krylov stein tails. Preprint and software repository, RH-64, 2026.